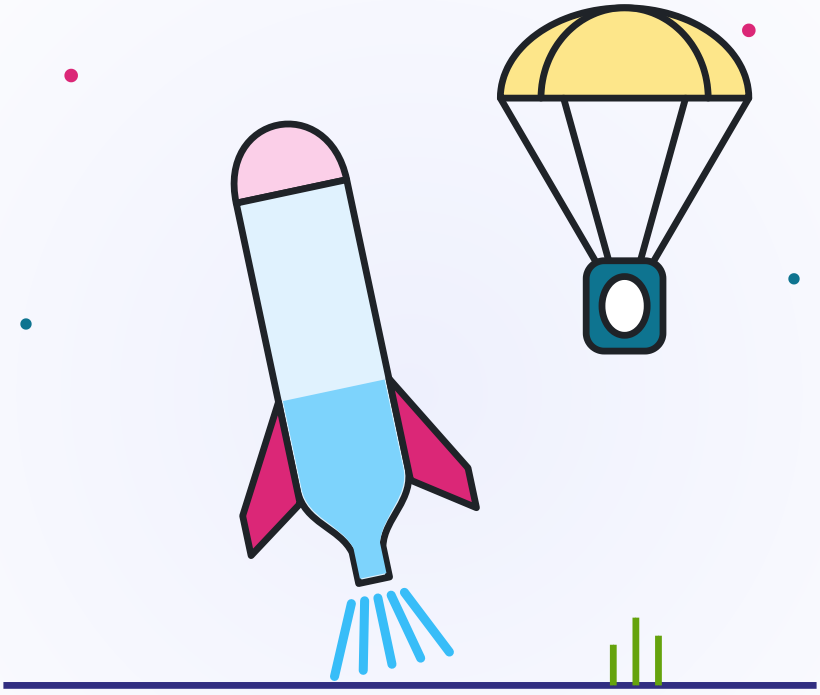
ENGINEER'S Logbook

Plan it. Build it. Launch it. Make it better.



ENGINEER ID

Name: _____

Partner: _____

Event: Egg-O-Naut

If found, please return to the DDMES Science Olympiad binder shelf.

How an engineer works

Real engineers keep a **logbook**. They write down every plan, every build, every launch, and every landing. This is yours. Bring it to every build session and keep it with your rocket at home.

1
PLAN
Draw your rocket on graph paper with real sizes.

2
BUILD
Tape only on the bottle. No metal anywhere.

3
LAUNCH
At team sessions only, with Z87+ glasses and the launch line.

4
LOG
Three timers, then write the middle time right away.

5
IMPROVE
Change ONE thing. Beat your score.

The Engineer's Code

- [ ] Z87+ safety glasses on to load, launch, and get the rocket.
- [ ] Stay behind the launch line. Never walk toward a pumped-up rocket.
- [ ] Go get the rocket only after 'All clear'.
- [ ] **Never launch at home.** Launches only happen with the team launcher.
- [ ] **You** build. Grown-ups watch and hand you tape.
- [ ] Change one thing at a time so you know what made the difference.

My build kit

- [ ] This logbook and a pencil
- [ ] A labeled 2 L carbonated soda bottle (keep the label!)
- [ ] Fin material: yard-sign plastic, file folders, or cardboard
- [ ] Duct tape and packing tape
- [ ] Cup and clay for a blunt nose
- [ ] Plastic bag and string for a parachute
- [ ] Z87+ safety glasses

Journal map

Know your event	p. 3-4
Safety Check and designs	p. 5-7
Stability, build, launch logs	p. 8-11
Graph and fix it	p. 12-13
At-home challenges	p. 14-16
Tournament, missions, badges	p. 17-19

Where engineers find ideas

Your launch data
Middle times show what really works.

The NCSO page
Rules, videos, and launch tips.

Maple seeds and leaves
Watch how spinning and flat shapes fall slowly.

Your landing spot
Did the egg part come off? Did the chute open?

Recycling bin
Bottles, cups, and plastic for new parts.

The score sheet
Check it before every launch day.

Did you know? A bottle rocket at 75 psi has air pushing on every square inch inside it with 75 pounds of force. That is why nobody walks toward a pumped-up rocket!

Know Your Event: Rules and Scoring

You will build a water rocket that carries a raw egg high into the air and brings it back down slowly and safely. The longer it stays up, the better, and you earn bonus seconds if the egg part separates and if the egg does not crack.

Rocket rules

Bottle	One plastic carbonated drink bottle, 2 L or less. Bring any label you take off.
Do not harm the bottle	No cutting, sanding, heat, hot glue, or super glue. Tape only on the bottle.
Fins	Every fin and part at least 5 cm above the bottle opening.
Nose	Blunt or round, never pointy. No metal anywhere.
Power	Water and air only, pumped to 75 psi.

How you score

Score = time aloft
+ 10 s if the egg part separates
+ 5 s if the egg survives
– 5 s for a non-safety rule break

Time aloft: three timers start when the rocket leaves the launcher and stop when the egg part touches the ground. The **middle** time counts.

Launch safety

- Z87+ safety glasses on to load, launch, and get the rocket.
- Stay behind the launch line.
- Never walk toward a pumped-up rocket.
- Go get the rocket only after 'All clear'.

Tournament routine

Bring	Up to 2 rockets, any removed labels, Z87+ glasses, your own funnel and measuring cup.
Water	About 1 minute per rocket to add water.
Launch	8 minutes to launch. One launch per rocket. Each rocket has its own parachute.

Practice: score it

1. Timers read 11.2 s, 11.6 s, and 11.4 s. What is the official time?

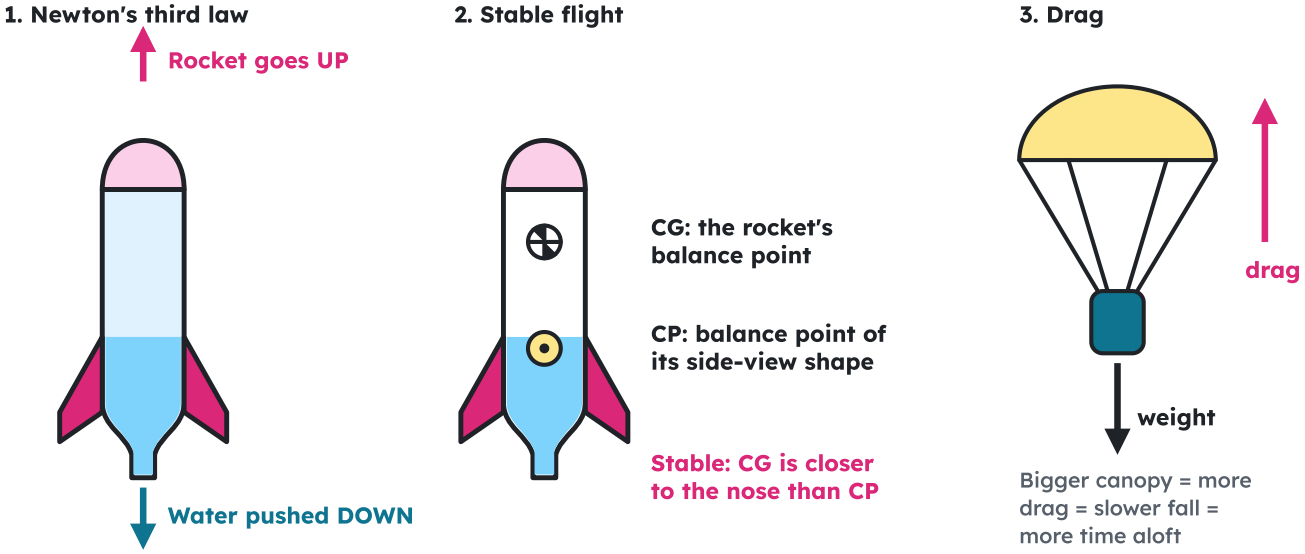
2. Time aloft 12.40 s. The egg part separated and the egg survived. No rule breaks. Score?

3. Time aloft 9.85 s. No separation, the egg cracked, and one non-safety rule break. Score?

Check: 1) 11.4 s (the middle one). 2) 12.40 + 10 + 5 = 27.40 s. 3) 9.85 + 0 + 0 − 5 = 4.85 s.

Know Your Event: Rocket Science

The rocket flies **upside down**: the bottle opening sits on the launcher, and the bottom of the bottle becomes the nose.



Up and stable

Newton's third law	Every push has an equal push back the other way.
Thrust	The push that moves the rocket. Water shoots down, the rocket goes up.
psi	Pounds per square inch: how hard the air pushes inside the bottle.
Center of gravity (CG)	The balance point of the rocket.
Center of pressure (CP)	The balance point of the rocket's side-view shape. Air pushes on the rocket around this spot.
Stable	CG is closer to the nose than CP, so it flies nose first.
To fix	Add clay to the nose (moves CG forward) or use bigger fins (moves CP back).

Bringing the egg down

Drag	Air pushing against a moving object. More drag means a slower fall and more time aloft.
Parachute	A big light canopy that makes lots of drag. Bigger canopy = slower fall.
Capsule	The part that holds and cushions the egg. Padding soaks up the bump.
Egg rules	The egg must come out easily. Nothing may be stuck to the egg.
Separation	If the egg part comes off the rocket, you get +10 s. It must be a different color from the rest of the rocket.
Recovery rules	Each rocket needs its own parachute. Springs or rubber bands must be relaxed at launch.

Quick check

The water is pushed _____, so the rocket goes _____. Adding clay to the nose moves the CG toward the _____.

Check: down; up; nose.

Safety Check

Check your rocket the same way the judges will, every launch day. Read each line with your partner and check the box only if the answer is YES. Fix a NO with tape if you can.

Check	Build session 2	Build session 3	Before tourney: Rocket 1	Before tourney: Rocket 2
SAFETY: any NO means that rocket does not launch				
2 L or less carbonated beverage bottle	☐	☐	☐	☐
Bottle label present	☐	☐	☐	☐
Only tape used on the bottle	☐	☐	☐	☐
Bottle not changed (no cuts, heat, glue, cracks)	☐	☐	☐	☐
No metal parts anywhere	☐	☐	☐	☐
Blunt or rounded nose	☐	☐	☐	☐
Recovery system is safe (if used)	☐	☐	☐	☐
Only water and air used to launch	☐	☐	☐	☐
Team wearing Z87+ safety glasses	☐	☐	☐	☐
NON-SAFETY: a NO costs 5 seconds				
1/2 inch PVC fits in the bottle opening	☐	☐	☐	☐
All parts 5 cm or higher above the opening	☐	☐	☐	☐
No store-bought rocket or toy parts	☐	☐	☐	☐
Egg is easy to remove and nothing is stuck to it	☐	☐	☐	☐
We did it with no outside help	☐	☐	☐	☐

Bottle cap test: set a bottle cap over the nose. The nose must not touch the inside top of the cap.

PVC test: a 1/2 inch PVC pipe must slide into the bottle opening loosely.

☐ Safety Officer badge: every line YES on each rocket before the tournament.

Design Sketch: Rocket 1

Draw Rocket 1 before you tape anything. Nose (the bottle's bottom) goes UP. Mark the 5 cm line above the opening, the fins, and where the egg rides.

Date _____ **Rocket** _____ **Bottle size** _____

SIDE VIEW: nose up. Mark CG and CP.

1 small square = 2 cm

RECOVERY: egg capsule and parachute

Part	Rule or tip	My plan
Fins: size and how many	every part 5 cm or more above the opening	
Nose	blunt or round; passes the bottle cap test	
Egg capsule color	different from the rocket (for the +10 s)	
Parachute size	from Lab I2: bigger = slower fall	
Water amount	from my fair test (p. 8)	

The ONE big idea in this version:

Design Sketch: Version 2

After your egg flight, redraw your rocket. Show your ONE change in a different color. Building a second rocket? It needs its own parachute and capsule.

Date _____ **Rocket** _____ **Bottle size** _____

SIDE VIEW: Version 2 or Rocket 2

1 small square = 2 cm

RECOVERY: egg capsule and parachute

Part	Rule or tip	My plan
Fins: size and how many	every part 5 cm or more above the opening	
Nose	blunt or round; passes the bottle cap test	
Egg capsule color	different from the rocket (for the +10 s)	
Parachute size	from Lab I2: bigger = slower fall	
Water amount	from my fair test (p. 8)	

The ONE big idea in this version:

_____

_____

Stability and Water Lab

Do this at Build session 2 with your real rocket and the 57 g egg stand-in in the nose.

1. Find CG and CP

Balance the rocket on one finger: that is the **CG**. Trace the rocket's side shape (with fins) on cardboard, cut it out, and balance it: that is the **CP**. Measure each one from the nose.

	CG: cm from nose	CP: cm from nose	CG closer to nose?	Swing test: nose first?
Rocket 1, first try				
Rocket 1, after my fix				
Version 2 or Rocket 2				

Not stable? Add clay to the nose (moves CG forward) or use bigger fins (moves CP back). My fix:

☐ Balance Master badge: CG is closer to the nose than CP, and the swing test flies nose first.

2. Water fair test (75 psi, same rocket, same egg stand-in)

Only the water changes. Three timers on every launch. Circle the middle time.

Water (mL)	Timer 1	Timer 2	Timer 3	Middle time
300				
650				
1,000				

My best water amount

_____ mL

Mark the fill line with a strip of tape on the outside of the bottle. Never marker or glue on the bottle.

What did we keep the same?

What surprised us?

☐ Water Wizard badge: fair test done and best water amount picked from our own data.

Sketch our rocket. Mark CG with ⊕ and CP with ●. Draw our tape fill line.

Build Log

Every time you build (at a session or at home), write one line. Tape only on the bottle, and **never launch at home**.

Date	Minutes	What I built or fixed	Rocket	Grown-up initials

☐ Rocket Builder badge: Rocket 1 drawn on p. 6 and built to match.

Launch Log

Launch #1

Date _____ **Rocket** _____ ☐ egg stand-in ☐ raw egg

What I changed (just ONE thing):

My prediction: it will stay up _____ seconds because…

Water (mL)	
Timer 1 / Timer 2 / Timer 3 (s)	
Middle time (s)	
Egg part separated? (+10 s)	
Egg survived, no wet spot? (+5 s)	
Non-safety rule break? (-5 s)	
Score (s)	

Draw the flight: up, parachute, landing

What happened? What I learned:

Next time I will change:

Launch #2

Date _____ **Rocket** _____ ☐ egg stand-in ☐ raw egg

What I changed (just ONE thing):

My prediction: it will stay up _____ seconds because…

Water (mL)	
Timer 1 / Timer 2 / Timer 3 (s)	
Middle time (s)	
Egg part separated? (+10 s)	
Egg survived, no wet spot? (+5 s)	
Non-safety rule break? (-5 s)	
Score (s)	

Draw the flight: up, parachute, landing

What happened? What I learned:

Next time I will change:

Launch Log

Launch #3

Date _____ **Rocket** _____ ☐ egg stand-in ☐ raw egg

What I changed (just ONE thing):

_____

_____

My prediction: it will stay up _____ seconds because…

_____

_____

Water (mL)	
Timer 1 / Timer 2 / Timer 3 (s)	
Middle time (s)	
Egg part separated? (+10 s)	
Egg survived, no wet spot? (+5 s)	
Non-safety rule break? (-5 s)	
Score (s)	

Draw the flight: up, parachute, landing

What happened? What I learned:

_____

_____

_____

Next time I will change:

_____

_____

Launch #4

Date _____ **Rocket** _____ ☐ egg stand-in ☐ raw egg

What I changed (just ONE thing):

_____

_____

My prediction: it will stay up _____ seconds because…

_____

_____

Water (mL)	
Timer 1 / Timer 2 / Timer 3 (s)	
Middle time (s)	
Egg part separated? (+10 s)	
Egg survived, no wet spot? (+5 s)	
Non-safety rule break? (-5 s)	
Score (s)	

Draw the flight: up, parachute, landing

What happened? What I learned:

_____

_____

_____

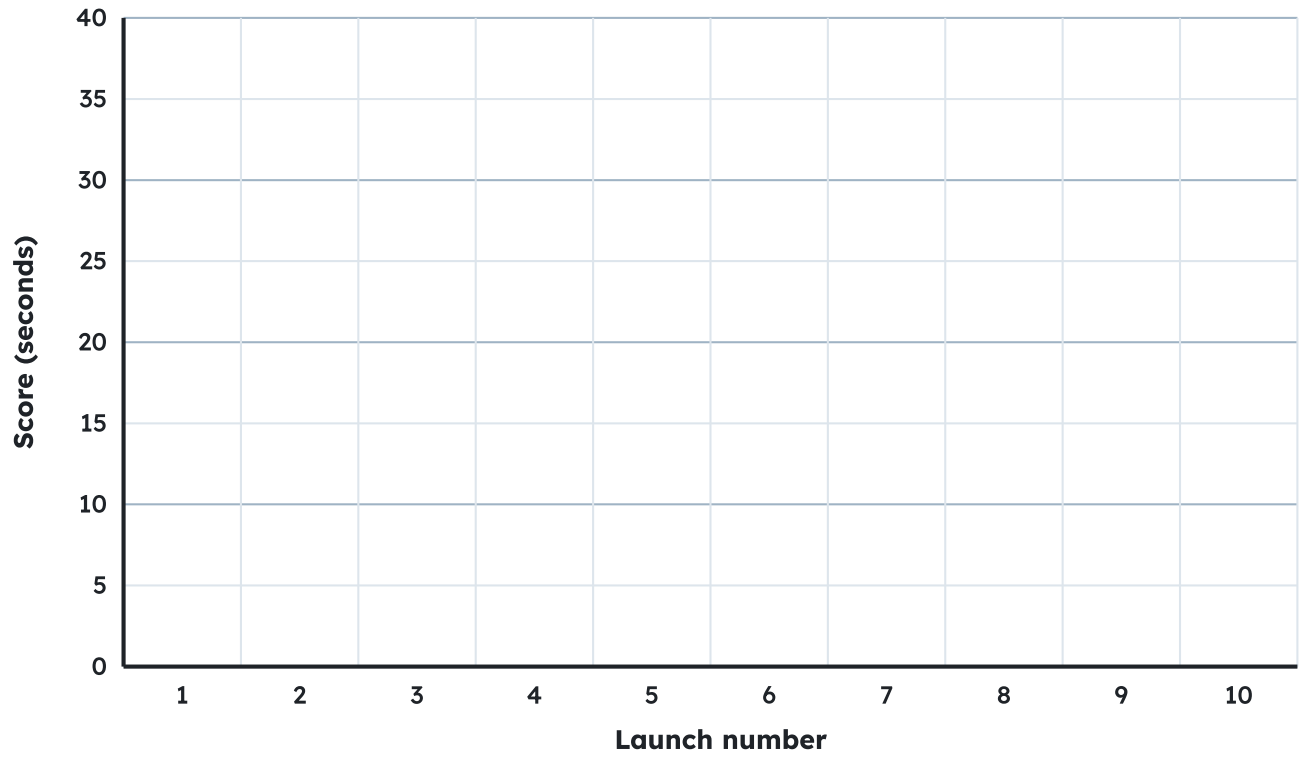
Next time I will change:

_____

_____

My Best-Score Graph

After every launch, put a dot at your score in seconds. Connect the dots. Is your line going up?



My best score

Score: _____ s Launch #: _____

Water: _____ mL Egg survived? _____

My goal line

Draw a dashed line across the graph at your goal.

Goal: _____ s

Separating (+10 s) and a safe egg (+5 s) add 15 seconds!

What change made my score go up the most?

☐ Rocket Upgrader badge: one change made, launched, and graphed.

What Went Wrong?

Find your problem. Read why it happens. Try the fix, then check the box. Engineers call this **troubleshooting**.

Problem	Why it happens	Try this	Tried
It tumbled or curved	The CG is too close to the CP (or behind it).	Add clay to the nose, or use bigger fins set farther back. Swing test again.	☐
It would not sit on the launcher	Fins or tape are below the 5 cm line.	Move every part to 5 cm or more above the opening.	☐
Short flight, low height	Too little or too much water.	Use the best amount from your fair test. Check the tape fill line.	☐
Parachute did not open	Packed too tight, or the lines tangled.	Fold it loosely. Lay the lines side by side. Practice packing it.	☐
Egg part did not separate	The capsule fits too tight on the rocket.	Loosen the fit so it slides off. Test by tipping the rocket over.	☐
Egg cracked	Too little padding, or it fell too fast.	Add padding around the egg. Use a bigger parachute.	☐
Fins bent or fell off	Weak tape or soft fin material.	Use yard-sign plastic and duct tape on both sides of the fin tab.	☐
It fell fast after the top	No parachute, or one too small.	Use your Lab I2 results: a heavier load needs a bigger canopy.	☐

My own problem

What happened:

Why I think it happened:

My fix, and did it work?

AT-HOME MINI BUILD CHALLENGE 1 · AFTER BUILD SESSION 1

Balloon Rocket Race

Earns the **Third Law Racer** badge. You build; a grown-up watches.

Goal: See Newton's third law in action: air rushes out one way, and the balloon zooms the other way.

You need

- [ ] Balloons
- [ ] A drinking straw
- [ ] About 5 m of string
- [ ] Tape
- [ ] 2 chairs
- [ ] Tape measure

Steps

1. Thread the string through the straw. Tie each end to a chair and pull the string tight.
2. Blow up a balloon with 5 breaths. Pinch the end shut (do not tie it).
3. Tape the balloon to the straw with the opening at the back. Let go!
4. Measure how far it went. Do 3 tries and use the middle distance.
5. Now try 10 breaths, then 15 breaths. Only the air changes.

Balloon	Try 1 (cm)	Try 2 (cm)	Try 3 (cm)	Middle (cm)
5 breaths				
10 breaths				
15 breaths				

Which way did the air go? Which way did the balloon go? Why?

How is your water rocket like this balloon?

Draw your build, or tape a photo here

AT-HOME MINI BUILD CHALLENGE 2 · AFTER BUILD SESSION 2

Paper Spinner Drop

Earns the **Drag Racer** badge. You build; a grown-up watches.

Goal: Make a paper spinner that falls as slowly as possible. More drag means more time in the air, just like your parachute.

You need

- [ ] Printer paper
- [ ] Scissors
- [ ] Tape
- [ ] Ruler
- [ ] A timer (a phone timer is fine)

Steps

1. Cut a strip 5 cm wide and 20 cm long.
2. Cut down the middle from the top, 8 cm long, to make 2 wings. Fold one wing forward and one back.
3. Fold the bottom up twice and tape it so it has a little weight.
4. Stand up, hold it as high as you can reach, and drop it. Time 3 drops and use the middle time.
5. Make a second spinner with 5 cm wings. Which one falls slower?

Spinner	Drop 1 (s)	Drop 2 (s)	Drop 3 (s)	Middle (s)
8 cm wings				
5 cm wings				
My own idea:				

Which spinner fell slowest? Use the word drag to explain why.

What did you keep the same so the test was fair?

Draw your build, or tape a photo here

AT-HOME MINI BUILD CHALLENGE 3 · AFTER BUILD SESSION 3

Egg Nest Drop

Earns the **Egg Guardian** badge. You build; a grown-up watches.

Goal: Build a capsule that keeps a raw egg safe in a drop. The egg must come out easily, with nothing stuck to it.

You need

- [ ] 1 raw egg in a zip bag
- [ ] Paper or plastic cups
- [ ] Padding: cotton balls, paper towels, bubble wrap, socks
- [ ] Tape
- [ ] Tape measure
- [ ] An old towel or a spot outside

Steps

1. Build a capsule around the bagged egg. Do not tape or glue the egg itself.
2. Drop it from 1 m onto the grass or the towel.
3. Open it and roll the egg on a paper towel. A wet spot means it cracked.
4. If it survived, go higher: 1.5 m, then 2 m (a grown-up holds it up high; nobody climbs).
5. If it cracked, change ONE thing and try again with a new egg.

Safety: wash your hands after touching raw egg. A grown-up does the high drop.

Drop height	My padding	Egg OK? (no wet spot)
1 m		
1.5 m		
2 m		

Which padding worked best? Why do you think so?

What will you change in your rocket's egg capsule?

Draw your build, or tape a photo here

Tournament-Day Checklist

Partners bring up to two water rockets they built before the tournament (plus any removed bottle labels and Z87+ safety glasses), pass a safety inspection, add water in about 1 minute per rocket, then have 8 minutes to launch each rocket once at 75 psi carrying a raw egg.

Pack the night before

- ☐ Rocket 1 (and Rocket 2, if we have one)
- ☐ A parachute and capsule for EACH rocket
- ☐ Any bottle labels we took off
- ☐ Z87+ safety glasses for both partners
- ☐ Our funnel and measuring cup
- ☐ This logbook

Our numbers

Rocket 1 water	_____ mL
Rocket 2 water	_____ mL
Parachute size	_____ cm
Safety Check (p. 5)	☐ all YES

Who does what

Adds the water	**Loads the egg**

At the launch

- ☐ Glasses on, both of us
- ☐ Pass the safety inspection
- ☐ Add water in about 1 minute
- ☐ Egg in, easy to take out, nothing stuck to it
- ☐ Parachute packed loosely
- ☐ Behind the launch line; wait for 'All clear'

My tournament result

Middle time: _____ s Separated? ___

Egg survived? ___ Score: _____ s

What I will tell next year's team

Our tournament rocket: draw it or tape a photo

Build Missions

One mission for each build session, plus one before the tournament. When you finish, show a coach or your Event Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
Build session 1 At home after	Balloon Rocket Race (p. 14).	Third Law Racer	☐	
Build session 1 At home after	Draw Rocket 1 (p. 6), build it, and log it (p. 9).	Rocket Builder	☐	
Build session 2 At the session	Find CG and CP, swing test, and water fair test (p. 8).	Balance Master / Water Wizard	☐	
Build session 2 At home after	Paper Spinner Drop (p. 15).	Drag Racer	☐	
Build session 3 At the session	Score your egg flight with 3 timers on a Launch Log (p. 10).	Flight Scorer	☐	
Build session 3 At home after	Egg Nest Drop (p. 16).	Egg Guardian	☐	
Before tournament Version 2	Safety Check all YES (p. 5). Draw Version 2 (p. 7) and graph your scores (p. 12).	Safety Officer / Rocket Upgrader	☐	

Tournament day: Sat Mar 13

- ☐ Rockets, each with its own parachute
- ☐ Bottle labels and Z87+ glasses
- ☐ Funnel, measuring cup, this logbook

- Run the Safety Check the night before.
- Know your water amount for each rocket.
- Nobody goes to a rocket before 'All clear'.

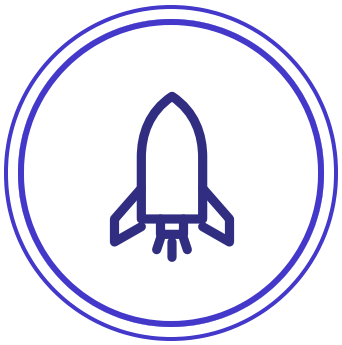
Engineer's notes: ideas, questions, and things to bring

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Egg-O-Naut**! Your Team Badge Passport has the same badges.



Third Law Racer
Balloon Rocket Race challenge
Initials: _____



Rocket Builder
Draw and build Rocket 1
Initials: _____



Balance Master
Find CG and CP; swing test
Initials: _____



Water Wizard
Water fair test
Initials: _____



Drag Racer
Paper Spinner Drop challenge
Initials: _____



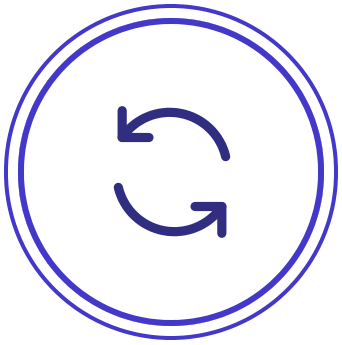
Egg Guardian
Egg Nest Drop challenge
Initials: _____



Flight Scorer
Score a flight with 3 timers
Initials: _____



Safety Officer
All YES on the Safety Check
Initials: _____



Rocket Upgrader
Make one change and graph it
Initials: _____